

Causal inference in latent space for time series and text data

IA pour la Science

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Introduction

We construct a Bayesian causal-inference model for multivariate time series. The goal is to identify a causal relationship between time series given the constructed model. We suppose that the unknown dynamics generated the observable time series from the latent space. Here we keep the model simple to expand it for 1) multimodal time series and control problems, 2) transformers and large models, 3) continuous-time dynamical systems and Neural ODE.

1 Discrete-time linear model

There given time series $\mathbf{x}_t$ as observations. The time series are generated by a discrete-time dynamical system in the latent space $\mathbf{z}_t$ with unknown dynamics:

$$\mathbf{x}_{t+1} = f(\mathbf{x}_t) + \boldsymbol{\zeta}_{t+1}, \quad (1)$$

where $\boldsymbol{\zeta}_t$ is the noise term. The linear model is

$$\mathbf{x}_t = \mathbf{W}\mathbf{x}_{t-1} + \mathbf{b} + \boldsymbol{\zeta}_t.$$

The time series are represented via *delay embedding*. The matrix of delay vectors for the scalar-valued series x_t is the Hankel matrix $\mathbf{X}$. Let the vector $\mathbf{x}_{t-1}$ be the first column of the matrix $\mathbf{X}_{t-1}$. Each column of the matrix $\mathbf{X}$ contains a point $\mathbf{x}_t$. Call *the phase trajectory* of a dynamical system the sequence $\{\mathbf{x}_t\}$. The dimensionality of the model is

$$\begin{pmatrix} w_{1,1} & w_{1,2} & \cdots & w_{1,N} \\ w_{2,1} & w_{2,2} & \cdots & w_{2,N} \\ \vdots & \vdots & w_{K,\tau} & \vdots \\ w_{K,1} & w_{K,2} & \cdots & w_{K,N} \end{pmatrix} \begin{pmatrix} x_{t-1} & x_{t-2} & \cdots & x_{t-(N+T)} \\ x_{t-2} & x_{t-3} & \cdots & x_{t-(N+T-1)} \\ \vdots & \vdots & x_{t-\tau} & \vdots \\ x_{t-N} & x_{t-(N-1)} & \cdots & x_T \end{pmatrix}.$$

Hankel matrix $\mathbf{X}_{t-1}$

The model parameters are

$$w_{\kappa,\tau}, \quad \tau = 1, \dots, K, \quad \kappa = 1, \dots, N.$$

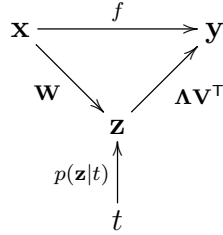
For the vector autoregression model $N = K$. Otherwise suppose $N \gg K$ and call N the original embedding dimension and K the reduced embedding dimension. The discrete time lag $\Delta t = 1$ and $T \ni \{1, \dots, t\}$ is the sample size. For d -variate time series $\mathbf{x}_t^{(1, \dots, d)}$, the matrix $\mathbf{X}$ is block-Hankel with d blocks. The matrix $\mathbf{W}$ in this case is $(K \times d \cdot N)$.

2 State space reconstruction

Introduce the singular value decomposition of the linear operator in the model (1):

$$f = \mathbf{W}\mathbf{\Lambda}\mathbf{V}^\top.$$

Denote by $\mathbf{z}$ the variable in the latent space. It corresponds to the dynamics of the system,



For simplicity omit the time t and the bias $\mathbf{b}$ in the model (1). Let the variable y be the forecast of the time series x_t . The variable $\mathbf{z}$ is the projection of the observable $\mathbf{x}$ onto the low-dimensional latent space.

In the terms of neural networks, this superposition is an encoder and a decoder. The linear operators $\mathbf{W}$ and $\mathbf{V}$ are the orthogonal and $\mathbf{\Lambda}$ is diagonal with positive descending elements. Each row of the matrix $\mathbf{W}$

$$\mathbf{W}\mathbf{x} = \begin{pmatrix} \mathbf{w}_1^\top \\ \mathbf{w}_2^\top \\ \vdots \\ \mathbf{w}_K^\top \end{pmatrix} \mathbf{x}$$

produces an element of a vector $\mathbf{z}$ in the latent space by the dot product

$$z_i = \mathbf{w}_i^\top \mathbf{x}.$$

Call the *embedding* the linear map $\mathbf{z} = \mathbf{W}\mathbf{x}$.

The system dynamics is represented by the generative model $p(\mathbf{z} | t)$. It approximates the embedding in the latent space with the support t .

3 Maximum a posteriori estimation of the parameters

The posterior distribution of the model parameters $\mathbf{w}$ given the data $\{\mathbf{x}, \mathbf{y}\}$ is

$$p(\mathbf{w}|\mathbf{x}, \mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{x}; \mathbf{w})p(\mathbf{w}|\mathbf{x})}{p(\mathbf{y}|\mathbf{x})}.$$

The likelihood function is

$$p(\mathbf{y}|\mathbf{x}; \mathbf{w}) = \prod_{t=1}^T p(\mathbf{y}_t|\mathbf{x}_t; \mathbf{w}).$$

The error given the posterior distribution of the parameters is

$$\mathcal{L}(\mathbf{w}) = \underbrace{\frac{1}{2}(\mathbf{f} - \mathbf{y})^\top \mathbf{B}^{-1}(\mathbf{f} - \mathbf{y})}_{\text{reconstruction error}} + \underbrace{\frac{1}{2}(\mathbf{w} - \boldsymbol{\mu})^\top \mathbf{A}^{-1}(\mathbf{w} - \boldsymbol{\mu})}_{\text{error in parameters}}.$$

Let the matrix $\mathbf{B} = \beta \mathbf{I}$ for the hypothesis that the variance β of the reconstruction error the same in the time segment $(1, \dots, T)$. The variance β is estimated as empirical distribution from the data over time t . The covariance $\mathbf{A}$ and the mean $\boldsymbol{\mu}$ of the normally-distributed parameters are estimated as empirical distribution from the data.

4 The criterion of causality

The criterion of the causal relationship between the elements of the model, we use the Kullback-Leibler divergence between the distributions of the parameters given the intervention $\text{do}(\mathbf{x})$:

$$\text{KL}(p(\mathbf{w}|\mathbf{x}, \mathbf{y}), || p(\mathbf{w}|\text{do}(X = \mathbf{x}_0), \mathbf{y})),$$

where $\{\mathbf{w}\} = \mathbf{W}$ is the set of the model parameters. The parameters are maximized according to the maximum a posteriori criterion.

For a linear operator $\mathbf{W}$, elimination of an edge from a node X , an element of the vector $\mathbf{x}_t$, is equivalent to zeroing the corresponding item $w_{\kappa, \tau}$. The node X corresponds to a column in $\mathbf{W}$. Since the parameter w is a random variable, the following statement holds, see Table 4.

Table 1: The action for a node based on the variance of the parameter w .

Expectation	Variance	Statement	Action
0	tends to 0	Sure that there is no impact	Remove the node
0	not tends to 0	Uncertain about the impact	Research to remove
not 0	tends to 0	Sure that there is an impact	Assert the node
not 0	not tends to 0	The impact must be specified	Research to assert
any	large values	Sure that there is no impact	Remove the node

5 The intervention algorithm

1. One step of back-prop in one layer.
2. Compute the covariance of the parameters (note the relation between Cov, Hessian and Fisher matrices).
3. Select a node X to make an intervention.
4. Make an intervention $\text{do}(X = \mathbf{x}_0)$ for a given node.
5. Compute the KL divergence between the distributions of the parameters with and without intervention.
6. Remove the intervention and go to step 1 until all nodes are processed.
7. The causal graph is constructed as a directed graph with edges weighted by the KL divergences.

6 $\text{do}(X)$ intervention policy

For regression or one-class classification, $\text{do}(X = x)$ is considered with values of x that lie outside the distribution of the time series at time t . For two-class classification, $\text{do}(X = x)$ is considered with values of x equal to the expectation of the opposite class.

What happens after the intervention? Until the forecasting and reconstruction error remains the same:

$$\mathcal{L}(p(\mathbf{y}|\mathbf{x}; \mathbf{w})) = \text{const},$$

1. Until the end of the short-term forecast horizon.
2. From one of sides, either X or Z , we start the intervention.
3. Start with the parameter of maximum variance of reconstruction error
4. Then, move to the parameters with lower variance.
5. Estimate the KL divergence of maximum posterior after each intervention.
6. Analyse the expectation and the variance of the parameters.
7. Eliminate the parameters with the highest KL divergence.
8. Keep the parameters with the lowest KL divergence.
9. Store them in the graph of causality.

Compare two graphs: direct and reverse.

7 Sequential causal inference

Results of the computational experiment: the norm of the gradient $\|\frac{\partial \mathcal{L}}{\partial \mathbf{W}}\|_2$ of the layer (l) causes the norm of the gradient in the previous layer $l - 1$

$$\left\| \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(l)}} \right\|_2 \rightsquigarrow \left\| \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(l-1)}} \right\|_2.$$

As well as the error on the training set causes the error on the validation set,

$$L_{\text{train}} \rightsquigarrow L_{\text{validation}}.$$

This holds for different optimization sessions.

The backprop procedure compute the parameters and their variance for sequence of layers. For several layers of a neural network

$$f = \sigma^{(L)} \circ \mathbf{W}^{(L)} \circ \sigma^{(l-1)} \circ \mathbf{W}^{(l-1)} \circ \dots \circ \sigma^{(1)} \circ \mathbf{W}^{(1)} \mathbf{x},$$

8 The Hessain, Fisher, and covariance matrices

The Hessian matrix of the loss function $\mathcal{L}$ is defined as

$$\mathbf{H} = \frac{\partial^2 \mathcal{L}}{\partial \mathbf{w} \partial \mathbf{w}^\top}.$$

Using the posterior of the parameters, we have

$$\mathbf{H} = \mathbb{E} \left[\frac{\partial^2 \log p(\mathbf{w}|\mathbf{x}, \mathbf{y})}{\partial \mathbf{w} \partial \mathbf{w}^\top} \right].$$

The Fisher information matrix is defined as

$$\mathbf{F} = \mathbb{E} \left[\left(\frac{\partial \log p(\mathbf{w}|\mathbf{x}, \mathbf{y})}{\partial \mathbf{w}} \right) \left(\frac{\partial \log p(\mathbf{w}|\mathbf{x}, \mathbf{y})}{\partial \mathbf{w}} \right)^\top \right].$$

Under certain regularity conditions, the following relation holds:

$$\mathbf{H} = -\mathbf{F}.$$

The covariance matrix of the parameters $\mathbf{w}$ is defined as

$$\mathbf{A} = \mathbb{E} \left[(\mathbf{w} - \mathbb{E}[\mathbf{w}])(\mathbf{w} - \mathbb{E}[\mathbf{w}])^\top \right].$$

Under certain assumptions, the following relation holds:

$$\mathbf{A} = \mathbf{F}^{-1}.$$

9 Computational experiment

For control problem we investigate a driven pendulum problem. A pendulum of a given length has its own period of oscillation. Driven by an external force a pendulum had forced oscillation. These two oscillations are simple and independent. They superposition are observed. A change in the frequency of the driving force disturbs the superposition. The goal is to identify the causal relationship between the driving force and the normal mode of the pendulum oscillations.

10 Appendix: Miscellaneous

Todo: Put plate notations for the bayesian model structure Γ in the slides (on monday). Introduce the bayesian structure of model Γ and associate it with the causal graph G . (The distribution of Γ could be coincided with the weights of the edges in G .) The weights of the arrows in the causation model means the posterior model structure Γ .

In the section 1: construct the manifold from the set of samples $\{\mathbf{z}\}$. Then control this manifold and the variables $do(Z = \mathbf{z})$ to infer the forecasting model f . Weight change and distribution as new data. Change parameters of the generative model $p(\mathbf{z}|t)$ and observe changes in the forecast. Use Variational Autoencoder instead of Normalising Flows. It simplifies the generative model.

Put the idea how to compute CI for continuous time dynamical system.

Discussion in January. Check the error: Infer another time series. It changes parameter, but does not changes the forecast. The error can stay the same.

Insert into section 2: Intervention in neuron weights versus intervention in the input data: proof of pros and cons.

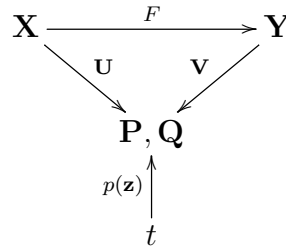
We apply $do()$ the sin to reconstruct the trajectory. We know the sin since we know the trajectory could be reduced to $\dim=2$.

Review: Hypothesis of *mélange* by M.S. Pinsker, or Ergodic Theory by Y.G. Sinai is exactly the Causal Inference.

Afterlife: When a Neural Network is a solution of the differential equation? Use the Neural ODE to test Causality in the mathematical models.

Domain transfer by causality graph: same model different datasets

11 Appendix: Control problem and Canonical Correlation Analysis



12 Appendix: continuous time dynamical system and delay embedding

For the continuous-time dynamical system, the state vector $\mathbf{z}(t)$ evolves according to the differential equation

$$\frac{d\mathbf{z}(t)}{dt} = f(\mathbf{z}(t), \mathbf{x}(t), \mathbf{y}(t)) + \boldsymbol{\varepsilon}(t),$$

and the observed time series are generated from the state vector via the observation functions:

$$\mathbf{x}(t) = h_x(\mathbf{z}(t)) + \boldsymbol{\xi}_x(t), \quad \mathbf{y}(t) = h_y(\mathbf{z}(t)) + \boldsymbol{\xi}_y(t),$$

where $\boldsymbol{\varepsilon}(t)$, $\boldsymbol{\xi}_x(t)$, and $\boldsymbol{\xi}_y(t)$ are noise terms.

The matrix of delay vectors for the scalar-valued series x_t is the Hankel matrix:

$$\mathbf{X} = \begin{pmatrix} x(t_1) & x(t_2) & \cdots & x(t_{N-T}) \\ x(t_1 - \Delta t) & x(t_2 - \Delta t) & \cdots & x(t_{N-T} - \Delta t) \\ \vdots & \vdots & \ddots & \vdots \\ x(t_1 - (T-1)\Delta t) & x(t_2 - (T-1)\Delta t) & \cdots & x(t_{N-T} - (T-1)\Delta t) \end{pmatrix},$$

where T is the length of the time series sample, $t \in (1, T)$. The time lag is Δt and N is the embedding dimension.

The continuous-time forecasting model is a convolution of two functions: the input signal $x(t)$ and the kernel $K(t)$ that represents the system dynamics. Thus, the output signal $y(t)$ can be expressed in the kernel terms:

$$y(t) = \int_{t-T}^t K(t - \tau)x(\tau)d\tau + \varepsilon(t).$$

The vector form of this equation is:

$$\mathbf{y}_t = \mathbf{K}^\top \mathbf{x}_t - \boldsymbol{\zeta}_t,$$